

G-02-epsilon0-import v1: audit ESS-2003 -i local concentration C_G

ZFL Lab

September 3, 2026

EXTERNAL-DEPENDENCY: ESS-2003 (Escauriaza-Seregin-Sverak) INPUT: C_{L^3} PROVEN:
 $T_* < \infty \implies \limsup \|u\|_3 = +\infty$ TARGET: $C_G: \exists t_n \uparrow T_*, x_n, r_n \downarrow 0 : \|u(t_n)\|_{L^3(B(x_n, r_n))} \geq \epsilon_0$

AUDIT:

1. What ESS-2003 actually proves: Theorem ESS: If u is suitable weak solution in $R^3 \times (-1, 0)$ with $\|u\|_{L_t^\infty L_x^3} \leq M < \infty$, then u is regular up to 0. Contrapositive gives C_{L^3} : $T_* < \infty \implies \limsup \|u\|_3 = \infty$. It does NOT directly state local ball lower bound with $r_n \rightarrow 0$.

2. Does ESS =i C_G ? ESS gives global blow-up criterion, not localized mass. To get C_G need additional concentration-compactness argument: Rescale near blow-up times, extract profile. If no concentration, scaling limit would be bounded L^3 solution that is regular, contradicting blow-up. This requires backward uniqueness + Carleman estimates already in ESS, but also profile decomposition lemma that is not stated as C_G in ESS paper.

Thus: ESS-2003 does NOT directly imply C_G as defined

3. What additional bridge is needed? - (i) Lemma: If $\|u(t_n)\|_{L^3(B(x_n, R))} < \epsilon_0$ for all small balls uniformly, then rescaled sequence converges to zero? Needs ϵ -regularity local version: There exists ϵ_* such that $\|u\|_{L_t^\infty L^3(B_1)} \leq \epsilon_* =_i$ regularity. This is related to Kozono-Taniuchi or ESS local version, not identical to global ESS. - (ii) Scale $r_n \downarrow 0$ guarantee: Requires $T_* - t_n \rightarrow 0$ and parabolic scaling $r_n \sim \sqrt{T_* - t_n}$ to keep critical norm invariant. If r_n does not go to 0, concentration is not critical.

4. Is ϵ_0 universal? ESS constant depends only on dimension, not on solution. If local version holds, ϵ_0 universal. But universality for C_G as defined requires audit of local ϵ -regularity constant, not global ESS constant alone.

5. Extra hypotheses: Suitable weak solution, finite energy, pressure in $L_{loc}^{3/2}$, etc.

STATUS:

C_G^{cond} NOT DERIVED from ESS-2003 alone

C_G remains OPEN, requires local ϵ -regularity + concentration compactness

ESS $\implies C_{L^3}$ PROVEN, but ESS $\not\Rightarrow C_G$ directly

Hence:

$C_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ remains untouched.

$C_G^{cond} = \text{OPEN/BLOCKED}$, dependency localized to local ϵ -regularity theorem (not global ESS).

No C_F . No contamination.

Next bridge $C_E^{weak} \wedge C_{L^3} \wedge C_G^{cond}$ cannot be attempted until local version audited as G-03.

FORBIDDEN respected: No claim “ESS =i C_G'' without explicit bridge.